

FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION

Book A Doctor – Online Doctor Appointment Booking System

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Classification

Project Documentation for Academic Submission

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1. Introduction

The healthcare sector increasingly relies on digital solutions to improve patient care and operational efficiency. Traditional appointment booking methods often involve long waiting times, manual record management, and communication delays.

Book A Doctor is a MERN Stack based healthcare appointment management platform designed to simplify the interaction between patients, doctors, and healthcare administrators.

The system enables patients to schedule appointments online, doctors to manage appointment requests, and administrators to oversee healthcare operations through a centralized platform.

2. Project Overview

Purpose

The purpose of this project is to provide a digital platform that simplifies healthcare appointment management while reducing manual effort and improving patient experience.

Target Users

Patients

- Register and login
- Search doctors
- Book appointments
- Track appointment status

Doctors

- Apply for doctor registration
- Manage appointments
- Approve or reject requests

Administrators

- Approve doctors
- Manage users

- Monitor system activities

Key Features

- Secure Authentication
 - Doctor Management
 - Appointment Scheduling
 - Notification System
 - Admin Dashboard
 - Doctor Dashboard
 - Responsive Design
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3. Problem Statement

Many hospitals and clinics continue to use manual appointment booking systems. These traditional methods create several challenges:

- Long waiting times
- Lack of appointment transparency
- Difficulty finding available doctors
- Poor communication
- Inefficient record management

The Book A Doctor platform addresses these challenges by providing a centralized digital healthcare appointment management solution.

4. Objectives

The objectives of the project are:

- Digitize healthcare appointment booking.
- Improve accessibility to healthcare services.
- Reduce administrative workload.
- Enhance communication between doctors and patients.
- Provide secure data management.
- Improve user satisfaction.

5. System Architecture

The system follows a three-tier MERN architecture.

Presentation Layer

React.js frontend providing user interfaces.

Business Logic Layer

Node.js and Express.js backend handling APIs and application logic.

Data Layer

MongoDB Atlas cloud database storing application data.

Architecture Flow:

Patient/Doctor/Admin

↓

React Frontend

↓

Node.js + Express Backend

↓

MongoDB Atlas Database

6. Technology Stack

Frontend:

- React.js
- Ant Design
- Redux Toolkit
- Axios

Backend:

- Node.js
- Express.js

Database:

- MongoDB Atlas

Authentication:

- JWT
- bcryptjs

Deployment:

- Vercel
- Render

Version Control:

- Git
 - GitHub
-

7. Setup Instructions

Prerequisites

- Node.js
- MongoDB Atlas Account
- Git
- VS Code

Installation

Clone Repository:

<https://github.com/nanda-dev7/VIP-C2-Book-A-Doctor-MERN>

Install Dependencies:

Frontend:

`npm install`

Backend:

`npm install`

Configure Environment Variables.

Run Frontend:

`npm run dev`

Run Backend:

npm run dev

8. Folder Structure

Client:

- components
- pages
- redux
- routes
- services

Server:

- config
- controllers
- middleware
- models
- routes

Each folder follows modular development principles for maintainability and scalability.

9. Application Workflow

Patient Workflow

Registration

→ Login

→ Search Doctor

→ Book Appointment

→ Receive Notification

Doctor Workflow

Apply Doctor

→ Admin Approval

→ Manage Appointments

→ Update Status

Admin Workflow

Review Doctor Applications

→ Approve/Reject

→ Monitor System

10. API Documentation

Authentication APIs:

- POST /register
- POST /login

Doctor APIs:

- POST /apply-doctor
- GET /get-all-doctors

Appointment APIs:

- POST /book-appointment
- GET /appointments

Admin APIs:

- GET /get-all-doctors
- POST /change-account-status

11. Authentication & Security

Security Measures:

- JWT Authentication
- Password Encryption (bcryptjs)
- Protected Routes
- Role-Based Access Control

Benefits:

- Prevents unauthorized access
 - Secures user credentials
 - Protects sensitive healthcare data
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12. User Interface

Screenshots Included:

- Login Page
- Registration Page
- Home Dashboard
- Doctor Dashboard
- Admin Dashboard
- Appointment Booking Page

The interface is designed to be responsive and user-friendly.

13. Testing

Testing Performed:

- Functional Testing
- User Acceptance Testing
- API Testing

Results:

All major functionalities passed successfully.

14. Deployment

Frontend:

Vercel

Backend:

Render

Database:

MongoDB Atlas

Benefits:

- Cloud hosting
 - High availability
 - Easy maintenance
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15. Known Issues

- Email notifications not implemented.
 - Online payment integration pending.
 - Mobile optimization can be enhanced.
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16. Future Enhancements

- Video Consultation
 - Online Payments
 - AI Doctor Recommendations
 - Mobile Application
 - SMS Notifications
 - Electronic Medical Records
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17. Conclusion

The Book A Doctor platform successfully demonstrates the implementation of a full-stack healthcare appointment management system using the MERN Stack. The application improves healthcare accessibility, reduces administrative effort, and provides a secure and user-friendly appointment booking experience.

The project meets its objectives and provides a strong foundation for future healthcare technology enhancements.

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